Cucurbit diseases- cucumber, melon, squash, pumpkins, and watermelon

1. Powdery mildew -
   *Erysiphe cichoracearum (*syn. *Golovinomyces cichoracearum*), *Sphaerotheca fuliginea (*syn. *Podosphaera xanthii)*
2. Downy mildew - *Pseudoperonospora cubensis*
3. Cercospora leaf spot - *Cercospora citrullina*, *C. melonis*, *C. lagenarium*
4. Fruit rot and cottony leak - *Pythium aphanidermatum*
5. Fusarium wilt - *Fusarium oxysporum* f. sp. *cucumerinum*, *melonis*, *niveum*, *lagenariae* and *momordicae*
6. Anthracnose - *Colletotrichum orbiculare*
7. Angular leaf spot - *Pseudomonas syringae* pv. *lachrymans*
8. Bacterial leaf spot - *Xanthomonas campestris* pv. *cucurbitae*
9. Mosaic - *Cucumber Mosaic Virus* (CMV), *Squash Mosaic Virus* (SqMV), and *Watermelon Mosaic Virus*-2 (WMV-2).

Cucurbit powdery mildew- ***Erysiphe cichoracearum*** (syn. *Golovinomyces cichoracearum*), ***Sphaerotheca fuliginea*** (syn. *Podosphaera xanthii*)

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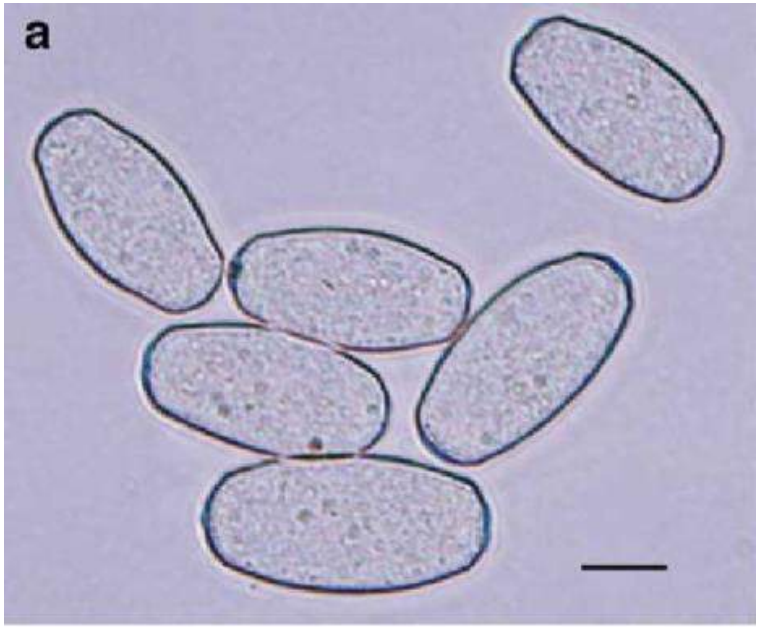
Cucurbit powdery mildew- *Erysiphe cichoracearum* (syn. *Golovinomyces cichoracearum*)**, *Sphaerotheca fuliginea*** (syn. *Podosphaera xanthii*)

Symptoms

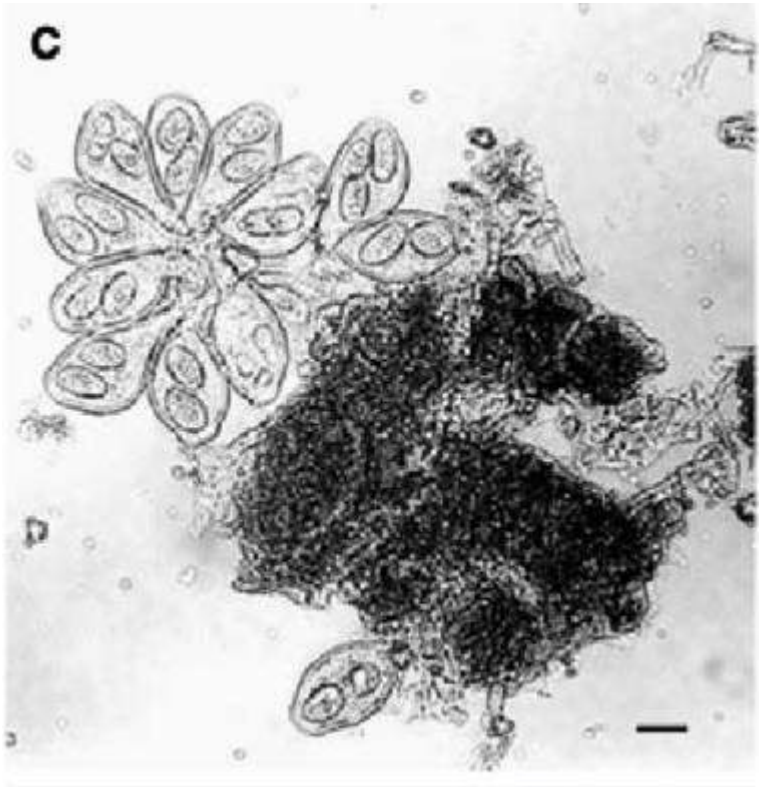
- Disease appear as white, floury coating on the leaves, stems and other succulent parts
- *S. fuliginea* is **more commonly reported worldwide**
- In *S. fuliginea* **the spots are dirty brown** in colour, **prefers warm weather**, while in *E. cichoracearum* spots are white in colour, prefers cooler weather
- This leads to premature defoliation and death
- Fruits remain undeveloped and deformed

Cucurbit powdery mildew- *Erysiphe cichoracearum* (syn. *Golovinomyces cichoracearum*)**, *Sphaerotheca fuliginea*** (syn. *Podosphaera xanthii*)



a. Conidia of *Golovinomyces cichoracearum*
b. Conidia of *Podosphaera xanthii*

Cucurbit powdery mildew- *Erysiphe cichoracearum* (syn. *Golovinomyces cichoracearum*)**, *Sphaerotheca fuliginea*** (syn. *Podosphaera xanthii*)



c. Cleistothecium with asci

d. Hyperparasitic fungus ***Ampelomyces quisqualis*** pycnidium on *Golovinomyces cichoracearum* basal cell of conidiophore

Cucurbit powdery mildew- *Erysiphe cichoracearum, Sphaerotheca fuliginea*

Disease cycle and epidemiology:

- The pathogens overwinter either in the form of **cleistothecia or conidia in cucurbitaceous weeds** in the neighbouring areas.
- **Moderate to warm temperature (25^0C)** coupled with high relative humidity and reduced sunshine hours favour the disease development

Cucurbit powdery mildew- *Erysiphe cichoracearum, Sphaerotheca fuliginea*

Management:

- **Do not spray sulphur fungicides as cucurbits are sensitive.**
- Collect and burn the infected plant debris.
- Spray the crop with fungicides like dinocap (0.06%), carbendazim (0.05%) or hexaconazole (0.05%) or difenoconazole (0.03%) and repeat at 10-14 days interval

Cucurbit downy mildew- *Pseudoperonospora cubensis*

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Cucurbit downy mildew- *Pseudoperonospora cubensis*

Symptoms

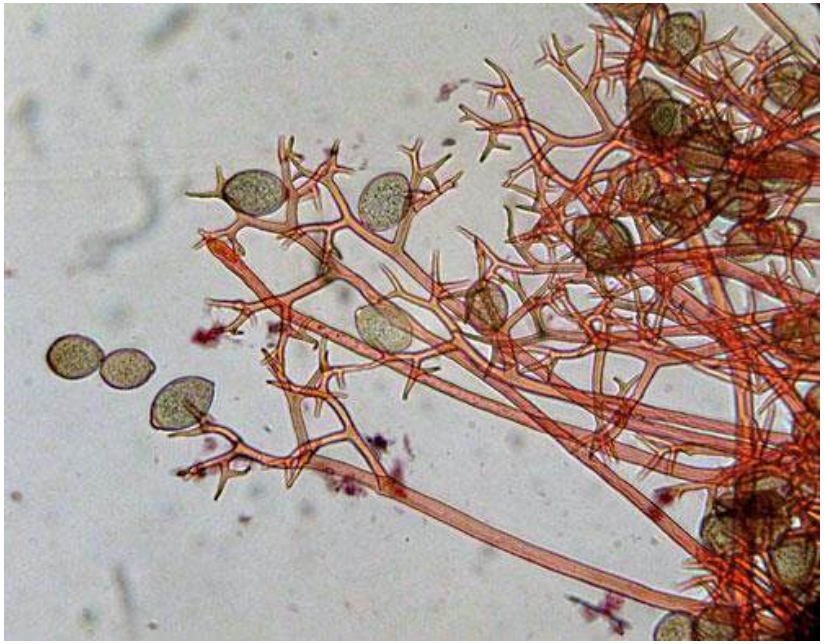
- **Pale green to yellow angular spots** restricted by veins on upper surface of leaves
- **Brownish downy growth** on the corresponding lower surface
- Later spots become **necrotic** with age
- Yellowing of leaves followed by **defoliation**
- Diseased plants get **stunted and die**
- Fruits produced **may not mature - poor taste**

Cucurbit downy mildew- *Pseudoperonospora cubensis*

Pathogen:

- The mycelium is **hyaline, coenocytic and intercellular.**
- Sporangiophores develop in groups of 1-5 and are **bulbous at base and branched dichotomously.**
- Sporangia form singly on **subacutely shaped sporuliferous tips,** are pale grey to purple in colour, ovoid to elliptical in shape, possess thin wall with a papilla at distal end.
- Sporangia germinate by producing zoospores.
- **Oospores are produced rarely** and are spherical, rarely obovoid to ellipsoid, light yellow to pale yellow with smooth wall

Cucurbit downy mildew- *Pseudoperonospora cubensis*



Sporangiophore bulbous at base and branched dichotomously subacutely

Sporangia

Cucurbit downy mildew- *Pseudoperonospora cubensis*

Disease cycle and epidemiology:

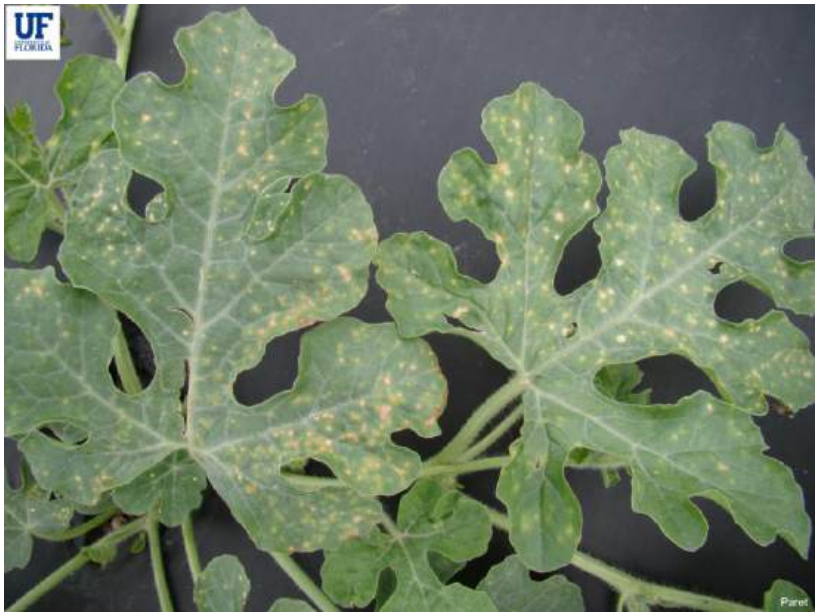
- The pathogen mainly **survives as mycelium and sporangiophores in living hosts during off-season.**
- **Survival through oospore is not common**.
- Moderate temperature (20-22° C) coupled with high rainfall and > 80 per cent RH favour disease development and spread.

Cucurbit downy mildew- *Pseudoperonospora cubensis*

Management:

- Destruction **of previous crop debris.**
- **Destruction of cucurbitaceous weeds** in and around
- Maintain proper plant density.
- Spray crop with mancozeb (0.25%) followed by one spray of metalaxyl + mancozeb or cymoxanil + mancozeb (0.25%) and two sprays of mancozeb (0.25%) at 10 days interval.

Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*



Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*

The center of the lesions dry out and has a white appearance. The lesions may coalesce to form blighted areas on the leaves.
Systemic yellowing may follow.

Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*

Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*

- Common on watermelon, muskmelon and cucumber

Symptoms

- **Minute water soaked spots** mostly on leaves
- Spots enlarge rapidly and becomes **circular to irregular with pale brown, tan or white centers and purple to almost black margins**
- The center of the lesions dry out **and has a white appearance**. The lesions may coalesce to form blighted areas on the leaves. Systemic yellowing may follow.
- Spots coalesce- **large blotches**
- Leaf dry, die - **scorched appearance**
- **Stems and fruits** are also infected

Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*

Symptoms contd……

- Severely affected plants have numerous leaf spots inside the canopy, plants turn **severely yellow and fruits** will be exposed to sun thus **sun scald on fruits**

Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*

Disease cycle

- **PSI:** Collateral hosts or plant debris
- **SSI**: Wind borne conidia

Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*

Cercospora spp. conidiophores and conidia (spores). The spores an easily move through wind and in free moisture.

The pathogen can survive on crop debris, volunteer plants and cucurbit weeds.

Cercospora leaf spot: *C. citrullina, C. melonis, C. lagenarium*

Management

- Maintain good soil drainage and good aeration between vines
- Two to three protective sprays with zineb@0.2%

Fruit rot or Cottony leak - *Pythium aphanidermatum, P. debaryanum*

Fruit rot or Cottony leak - *Pythium aphanidermatum, P. debaryanum*

- **Serious disease** during transit, storage and marketing
- **Bruises on the fruit** - aggravate the disease as many weak parasites are involved.
- Commonly occurring under humid conditions during rains

Symptoms

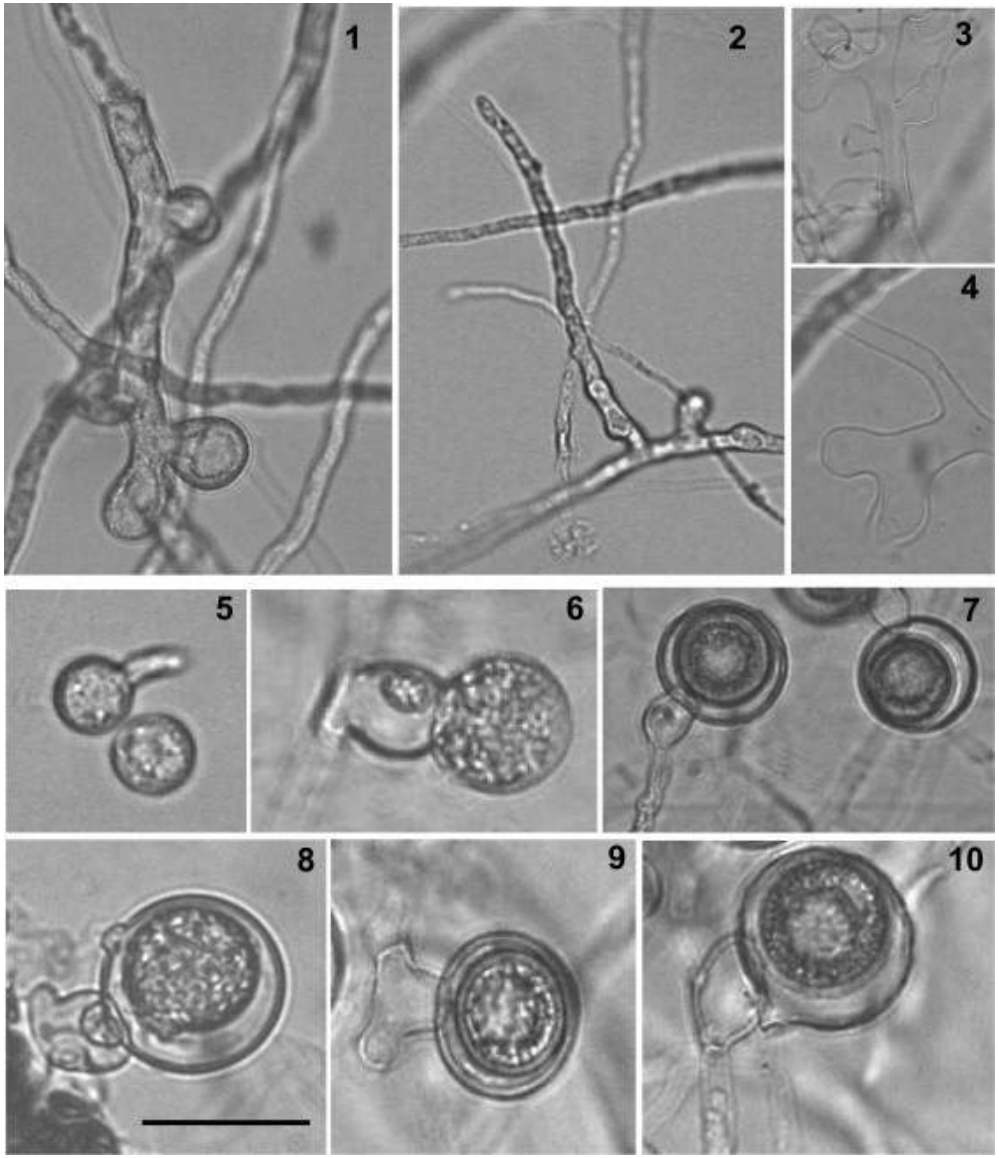
- Fruits, contact with soil affected much.
- On skin of the fruit initially shows **soft, dark green water soaked lesions** which gradually develop into a watery soft rot.

Fruit rot or Cottony leak - *Pythium aphanidermatum*

Symptoms contd…

- On **rotting portion, luxuriant cottony mycelium** develops abundantly and appears as **if wrapped in cotton**
- On the margin of the cottony growth, the skin of the fruit looks **dark green and water soaked** without mycelia growth indicates killing in advance activity of the pathogen.
- In watermelon decay starts at the blossom end

Fruit rot or Cottony leak - *Pythium aphanidermatum*



1,2. Zoosporangium.
3,4. Empty zoosporangia.
5. Germinating zoospores.
6,7 Young oogonium and intercalary antheridium.
8,9 Intercalary antheridia.
10. Oospore.

Fruit rot or Cottony leak - *Pythium aphanidermatum*

Disease cycle

- **Soil inhabitant** as saprophyte on dead organic matter
- High temp., high rainfall/excessive irrigation in hot weather

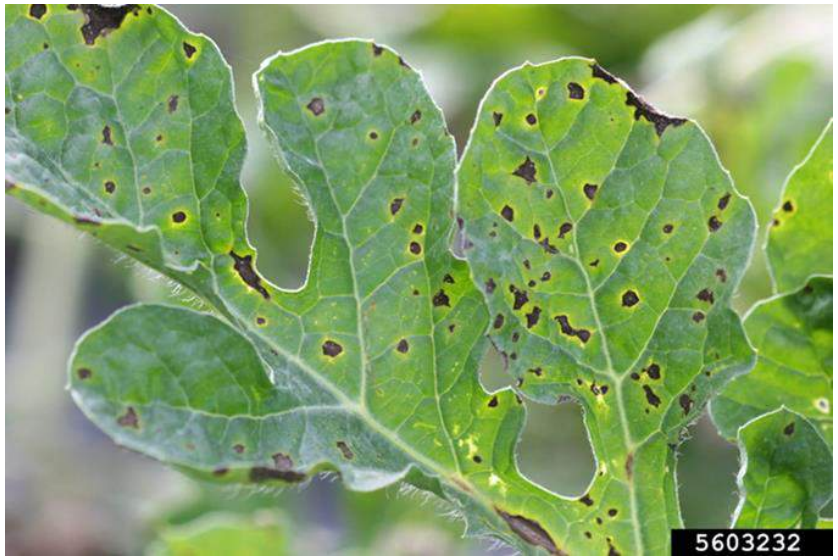
Management

- Providing good drainage
- Avoidance of excess irrigation
- Crop rotation
- Preventing the contact of mature fruits with soil by wrapping in cucumber leaves
- Seed treatment chloroneb, mancozeb, thiram, captan
- Soil drenching with copper oxychloride

Cucurbit Anthracnose-*Colletotrichum orbiculare*

Cucumber

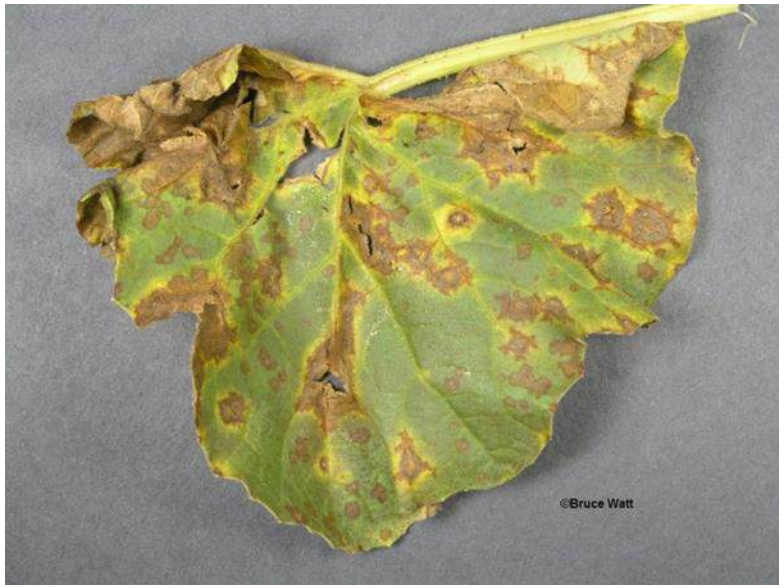
Cucurbit Anthracnose-*Colletotrichum orbiculare*



Watermelon

Cucurbit Anthracnose-*Colletotrichum orbiculare*

Cucurbit Anthracnose-*Colletotrichum orbiculare*



Cucurbit Anthracnose-*Colletotrichum orbiculare*

Cucurbit Anthracnose-*Colletotrichum orbiculare*

Water melon, musk melon, cucumber and bottle gourd

Symptoms

- Lesions on seedlings, leaves, petioles, stems and fruits
- Symptoms vary with host and part affected
- Lesions appear near veins, rough, light brown to dark brown in colour may reach 1 cm in dia.
- Necrotic part of the lesion may fall off - **shot hole**
- Lesions on **stems and petioles** are shallow and elongated
- **On young fruits - black spots, malformed and aborted**

Cucurbit Anthracnose-*Colletotrichum orbiculare*

Symptoms cond…

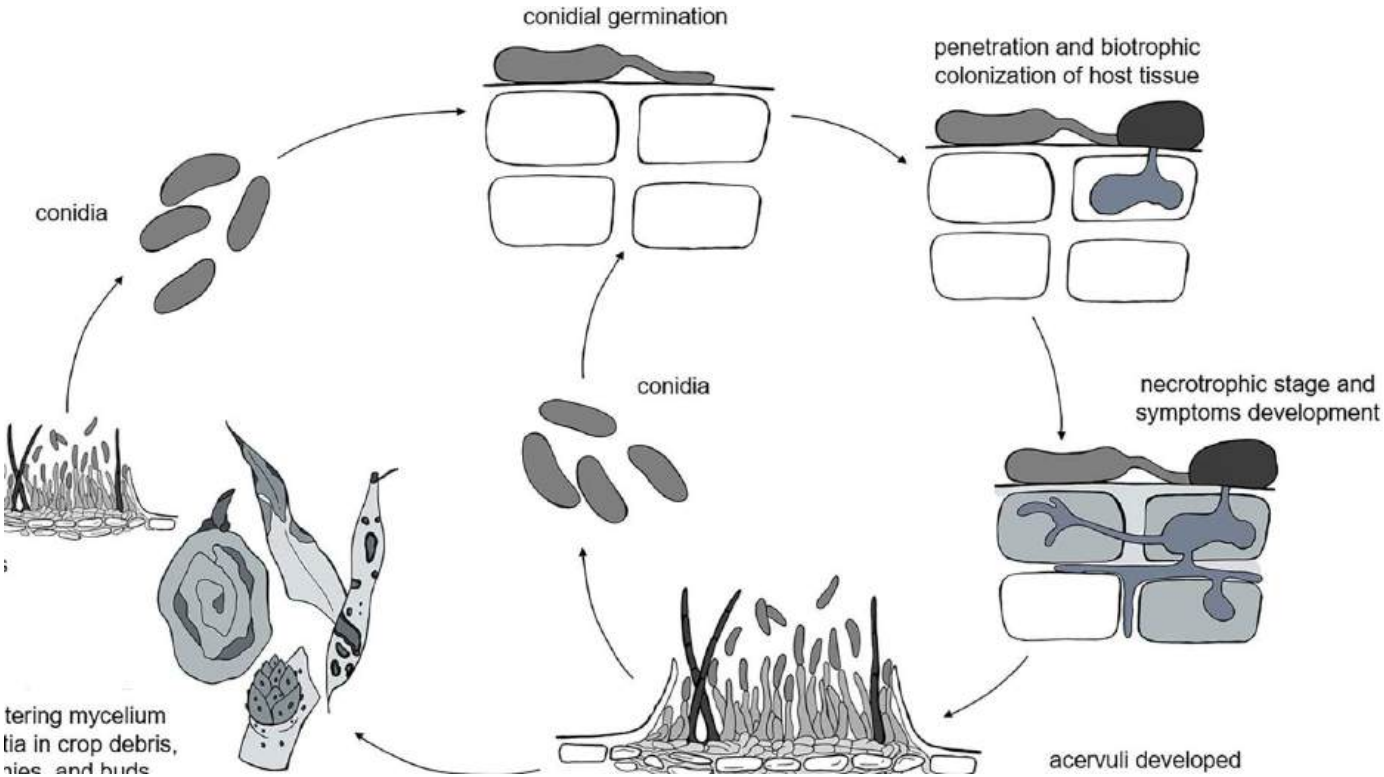
- **On fruits** - circular, sunken, water soaked spots covering larger area
- In **watermelon** irregular lesions, brown to black
- **Old spots turn black and are covered with pink spore** masses under moist weather.

Cucurbit Anthracnose -*Colletotrichum orbiculare*

Pathogen:

- The disease is caused by *Colletotrichum orbiculare* PS: *Glomerella lagenarium*)
- The fungus produces black stromata, bearing black setae and hyaline conidiophores.
- Conidia are produced on conidiophores by budding and are one celled, hyaline, oblong to ovate.

Cucurbit Anthracnose-*Colletotrichum orbiculare*



Cucurbit Anthracnose - *Colletotrichum orbiculare*

Disease cycle and epidemiology:

- The fungus is both soil and seed borne.
- Cucurbitaceous weed hosts serve as source of perennation.
- Conidia are disseminated by moist wind or rain splashes or carried through implements.
- Spore germination is optimum at 22-27° C temperature and 100 per cent RH.

Cucurbit Anthracnose - *Colletotrichum orbiculare*

- Infected plant debris, volunteer plants, alternate hosts-snake gourd, ash gourd, long melon, round melon

Management

- Crop rotation with non cucurbitaceous for one year
- Destruction of cucurbitaceous weed hosts
- Deep ploughing of crop residues immediately after harvest.
- Use healthy seed and treat seed with carbendazim (0.2%) Spray the crop with carbendazim (0.1%) or mancozeb (0.25%) or their combination at 10-14 days interval

Cucurbit Fusarium wilt - *Fusarium oxysporum* f. spp

Cucurbit Fusarium wilt - *Fusarium oxysporum* f.spp

Bacterial Wilt

Fusarium Wilt

Cucurbit Fusarium wilt -

Fusarium oxysporum f. sp. *cucumerinum* - cucumber
melonis - muskmelon
niveum - water melon
lagenariae - bottle gourd
momordicae - bitter gourd

Symptoms

- In affected plants wilting **mostly** appear at **flowering and fruiting stage**
- Flaccidity of leaves, yellowing and marginal necrosis
- Basal leaves **loose turgidity first proceed upward**
- Wilting may be sudden/if gradual take **4-5 days for symptom development**

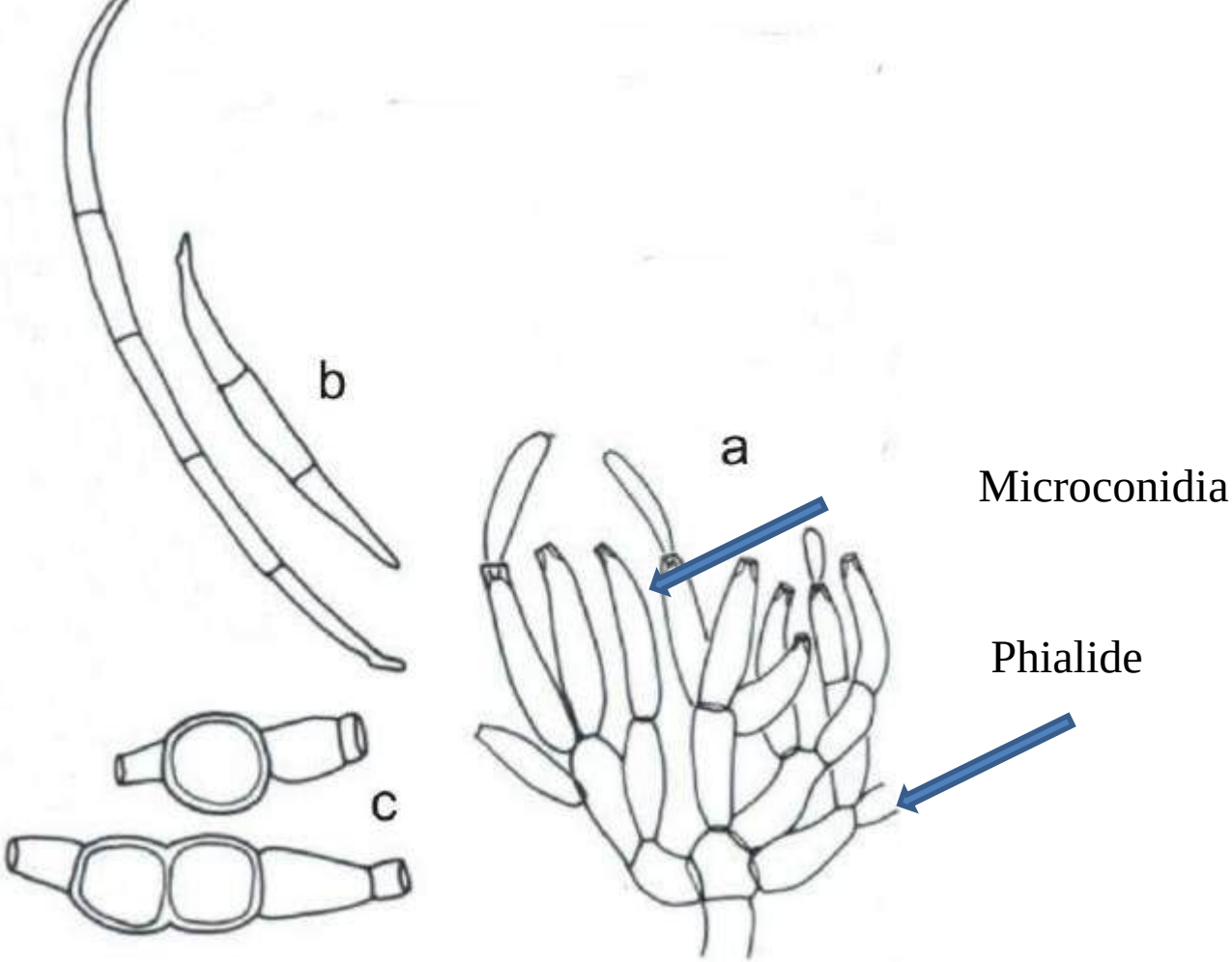
Cucurbit Fusarium wilt - *Fusarium oxysporum* f. spp

Symptoms contd…

- **Striking symptom -lateral roots** replace the tap root
- Vascular discolouration may or may not be there
- Plant may remain retarded without fruiting

Pathogen:

- The fungus produces septate mycelium and micro and macro conidia.
- Chlamydospores are also produced.



Three different spores produced by *Fusarium oxysporum* .
a: Sporodochium; b: Macroconidia; c: Chlamydospores .

Cucurbit Fusarium wilt - *Fusarium oxysporum* f.spp

Disease cycle and epidemiology:

- The fungus is **seed borne** (Bottle gourd, cucumber, musk melon, pumpkin) and **soil borne**
- Survives as chlamydospores.
- A temperature between 18-22° C, high light intensity and relative humidity favour the disease development.
- High nitrogen and low potassium enhance disease incidence.

Cucurbit Fusarium wilt - *Fusarium oxysporum* f.spp

Management:

- Follow crop rotation with non cucurbitaceous crops garlic, raddish, onion, beet root, lucerne.
- Soil solarization for 40-45 days during summer months.
- Apply soil amendments like neem and mustard cake (250 g/m2).
- **Cultivation of resistant varieties**

  Musk melon - Durgapura Madhu, Punjab Sunehri

  Melon- MR-1

Cucurbit Fusarium wilt - *Fusarium oxysporum* f.spp

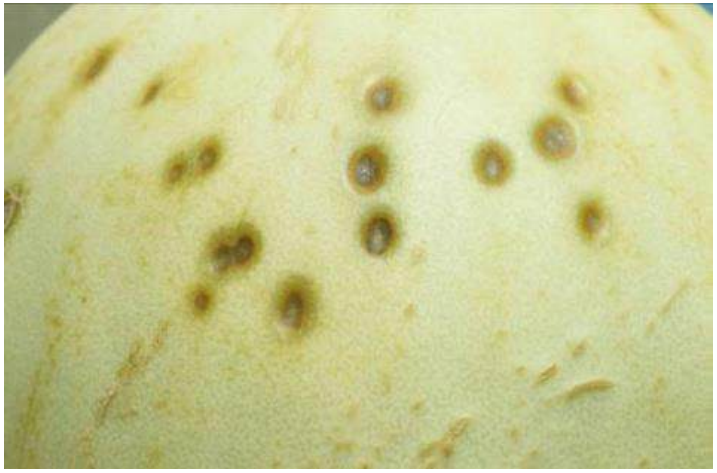
Management

- Drench the plants with carbendazim (0.1%).

- Application of non-pathogenic isolates of *Fusarium* or *Pseudomonas putida* as bioagents.

Angular leaf spot – *Pseudomonas syringae* pv. *lachrymans*

Angular leaf spot – *Pseudomonas syringae* pv. *lachrymans*



Angular leaf spot – *Pseudomonas syringae* pv. *lachrymans*

Symptoms:

- Symptoms appears on aerial parts *viz*., leaves, petioles, stems and fruits.
- On leaves the disease first appears as small, water-soaked spots which enlarge to about 3 mm in diameter without yellow halo.
- The spots become tan on the upper surface and gummy or shiny on the lower surface due to drying of bacterial ooze
- Foliar lesions become angular as they grow (vein limited), and tan to brown in colour when mature.

Angular leaf spot – *Pseudomonas syringae* pv. *lachrymans*

Symptoms contd…

- Mature lesions may tear apart and drop, leaving a hole
- Fruit lesions are usually circular and smaller than foliar lesions, and can either cause fruit drop, deformation, or superficial whitish cracked lesions.
- Internal rot may develop when secondary pathogens enter through the cracked lesion surface.
- Under humid conditions, bacteria ooze from the lesions as a milky exudates that dries and forms a whitish crust.

Angular leaf spot – *Pseudomonas syringae* pv. *lachrymans*

Pathogen:

- The disease is caused by bacterium *Pseudomonas syringae* pv. *lachrymans*
- It is Gram -ve, aerobic, non spore forming, straight rods, motile with 1-5 polar flagella.

Disease cycle and epidemiology

- The bacterium overwinters in infected seed as well as in plant debris.
- Optimum temperature for disease development is 20-24° C and moisture is most important factor for disease development and spread.

Angular leaf spot – *Pseudomonas syringae* pv. *lachrymans*

Management:

- Follow crop rotation.
- Deep ploughing , collection and destruction of plant debris.
- Use disease-free seed produced in dry areas and soak them in Streptocycline (100 ppm) for 30 minutes and dry under shade before sowing.
- Balanced nitrogen fertilisers application.
- Do not work in the crop when it is wet.
- After symptom appearance spraying with Streptocycline (100 ppm) or COC (0.3%) at 10 to 14 days interval.

Bacterial leaf spot – *Xanthomonas campestris pv. cucurbitae*

Bacterial leaf spot – *Xanthomonas campestris pv. cucurbitae*

Bacterial leaf spot – *Xanthomonas campestris pv. cucurbitae*

Symptoms

- Symptoms may appear on both the foliage and the fruit
- On the foliage, the disease causes small somewhat round water-soaked or greasy lesions on the underside of the leaf.
- On the upper leaf surface, the spots turn brown with a distinct yellow halo
- As spots, enlarge, they eventually become delineated by veins resulting in angular lesions, thin, brown, translucent centres and yellow halo.
- Initial lesions are typically small, round, slightly sunken with a tan centre

Bacterial leaf spot – *Xanthomonas campestris pv. cucurbitae*

Symptoms contd…

- As the spots enlarge (reaching up to 15 mm in diameter), they become noticeably sunken and the rind may crack.
- **Rarely** – infect young stems, petioles, and young fruits
- **On fruits** small, water-soaked areas with **light brown ooze**
- Infection extends into the seed cavity of the fruit results in **seed infection**.
- The flesh rots and seeds may become contaminated with the bacterium.

Bacterial leaf spot – *Xanthomonas campestris pv. cucurbitae*

- Seed and infected crop residue, as well as rain splash and the movement of people and machinery.

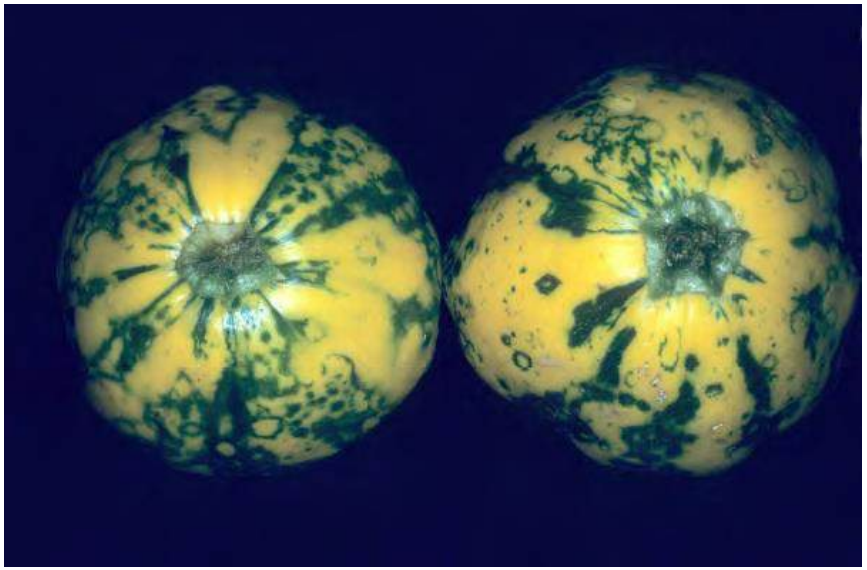
Management

- Use disease-free seed.
- Do not work in the crop when it is wet.
- Crop rotation
- Destruction of plant debris
- Deep ploughing
- Chemical treatment

Cucumber mosaic- CMV, SqMV, WMV-2

Cucumber mosaic- CMV, SqMV, WMV-2

Cucumber mosaic- CMV, SqMV, WMV-2



Cucumber mosaic- CMV, SqMV, WMV-2

70 different viruses were reported to cause a wide range of diseases in cucurbits in the genera *Begomovirus*, *Potyvirus*, *Cucumovirus*, *Tobamovirus*, *Tymovirus*, *Nepovirus*, and *Polerovirus* across the world.

In India - dominance of viruses- Begomovirus (93%), Potyvirus (46%), *Cucumber green mottle mosaic virus*- 39%, Polerovirus (9%), Orthotospovirus (2%)

Six viruses associated with cucurbit crops have been documented from Southern India during 2018 published in 2021 .

Tomato leaf curl New Delhi virus,
Squash leaf curl China virus
Papaya ring spot virus (PRSV)
Cucumber green mottle mosaic virus (CGMMV)
Zucchini yellow mosaic virus (ZYMV),
Cucumber mosaic virus (CMV-2%)

Cucumber mosaic- CMV, SqMV, WMV-2

- Of the many viruses that attack cucurbits, three commonly found viruses are C*ucumber Mosaic Virus (CMV)*, *Squash Mosaic Virus (SqMV)*, and W*atermelon Mosaic Vi*rus-2 *(WMV-2)*.

- These viruses differ in their host range, method of transmission.
- Symptoms produced by these viruses are similar.

Symptoms:

- When plants become infected in the 6 to 8 leaf stage, symptoms first appear on the youngest, still expanding leaves.

Cucumber mosaic- CMV, SqMV, WMV-2

Symptoms contd..

- A mosaic pattern develops-healthy, dark-green leaf tissue intermingled with light- green and yellow tissue
- Leaves are often distorted, crinkled, curled, and stunted.
- Vines may appear bunchy because of the shortening of the internodes.
- In severe cases, older leaves may die.

Cucumber mosaic- CMV, SqMV, WMV-2

Symptoms contd..

- Few fruit set on plants infected early in the growing season.
- Cucumber fruit may show yellow and green mottling or have dark green **"warts"** on pale green fruit.
- In sometimes smooth and pale whitish green (called "white pickle") and more blunted at the ends than fruit produced on healthy vines.
- Fruits are often misshapen, mottled, warty and reduced in size
- Late infections - bumpy or deformed fruits

Cucumber mosaic- CMV, SqMV, WMV-2

Pathogen:

- CMV which belongs to genus Cucumovirus.
- The particle size of virus is 29nm.
- SqMV which belongs to genus Comovirus
- WMV-2 which belongs to genus Potyvirus
- PRSV

Cucumber mosaic- CMV, SqMV, WMV-2

Disease cycle and epidemiology:

- With the exception of SqMV, which is seedborne in melon and transmitted by beetles, the other major viruses are transmitted by several aphid species in a nonpersistent manner.
- CMV has wide host range several weeds and cultivated hosts- vegetables, ornamentals including members of cucurbitaceae, solanaceae, malvaceae, brassicaceae and leguminosae.
- It is transmitted in non persistent manner by aphids like *Myzus persicae* and *Aphis gossypii*.
- The virus is also known to be transmitted mechanically and through seed.

Cucumber mosaic- CMV, SqMV, WMV-2

Management

- Rogue out infected plants
- Eliminate perennial weed hosts
- Rest. Source Cucumber – Chinese cv.
  Melons –Japanese, Korean and Chinese
- Planting barrier crops, applying sticky traps or covering the ground with an aphid deterrent material like aluminium foil boards can help in reducing the disease incidence.
- Use of virus free seed together with erradication of virus reservoirs are effective.
- Vector control - monocrotophos (1.5 ml) or malathion (2 ml) or dimethoate (1ml) or phosphamidon (1.5 ml) / lt